

MATTEO MARCHETTI

Software Engineer

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SUMMARY

Software engineer with three years in backend and enterprise systems, today building .NET microservices at TD Group. In my own time I write neural networks from scratch in NumPy, from the autograd engine up to a small GPT-style model, and I am moving toward machine learning engineering.

WORK EXPERIENCE

Software Engineer

Sep 2025 - Present

TD Group - Pisa, Italy

- Architected a local cache layer for .NET microservices serving static data, cutting database load and response times by 50%.
- Replaced an inaccessible Keycloak setup with a custom authorization layer for app-to-service communication.
- Owned an email service now carrying 10,000+ transactional messages per month across the platform.
- Shipped a configuration microservice consumed by every other service in the project.
- Delivered a Profile microservice and managed the integration work with the front-end developer. Serving 5000 request per day.
- Led the refactoring of GoCommons, a shared Go library used by 15+ microservices, closing bugs and security holes while lowering runtime memory use.
- Brought Entity Framework into the stack, wrote the internal guide and trained the team on SQL injection prevention.
- Standardized a .NET microservice template on Controller-Service-Repository and Unit of Work, now the base of every service.
- Set up the first CI/CD pipelines and DevOps conventions for the engineering workflow.

Software Developer

Jan 2024 - Aug 2025

E80 Group - Reggio Emilia, Italy

- Wrote C# customizations on top of the standard product so each Industry 4.0 plant matched the end client's layout and process rules.
- Worked at the boundary between high-level software and PLC integration, tracing material-handling logic across both layers.

Software Developer

Jan 2023 - Jan 2024

Abstract - Lucca, Italy

- Built responsive single-page interfaces and the REST endpoints behind them in a microservices-oriented architecture.
- Covered backend controllers and endpoints with unit tests.
- Kept legacy enterprise applications alive in production, fixing defects and adding features.
- Wrote and maintained stored procedures on Microsoft SQL Server and IBM DB2.
- Moved data from DB2 to SQL Server through SSIS packages.

PERSONAL PROJECTS

Thorcino | github.com/cecinuga/thorcino

- A PyTorch-inspired deep learning framework on NumPy: automatic differentiation engine, layers, loss functions, DataLoader and the full training lifecycle.
- Added SGD and Adam optimizers, deriving backpropagation and gradient computation by hand instead of trusting a library.

Tiny LM | github.com/cecinuga/tiny-lm

- A small GPT-2-like language model trained from scratch.

Fuzzy | github.com/cecinuga/fuzzy

- A concurrent fuzzing tool: parallel execution and payload reuse keep throughput high and avoid repeated work.

ArXiv MCP Server | github.com/cecinuga/arxiv-mcp-server

- An MCP server that lets LLM agents query ArXiv; my way into the protocol and into daily agent workflows.

Lacli | github.com/cecinuga/lacli

- A command-line linear algebra tool for matrix and vector operations over standard input/output and files.

SKILLS

- **Machine Learning / Deep Learning:** Neural networks, LLMs, AI agents, MCP, NumPy, PyTorch
- **Languages:** Python, C#, Go
- **Frameworks:** .NET, Entity Framework, ASP.NET Core
- **Data & Messaging:** SQL Server, IBM DB2, Redis, RabbitMQ, SSIS
- **Architecture & DevOps:** Microservices, distributed systems, CI/CD, Controller-Service-Repository, Unit of Work, CQRS, Message-Driven

LANGUAGES

- **Italian:** Native | **English:** B1, intermediate